



University of Colorado  
Denver

Department of Computer Science & Engineering

CSCI/CSCY 4407: Security & Cryptography

Assignment 1: [Quarter-Round Function Analysis & Implementation](#)

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**Spring 2026**

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## Assignment Instructions

This is an **individual assignment**. Each student is required to complete and submit their **own original work**. You may discuss general cryptography concepts with classmates; however, **you are not permitted to share, reuse, or copy** solutions, calculations, code, scripts, screenshots, written explanations, experimental results, or final answers from other students.

For this assignment, you will analyze and implement components of the **ChaCha20 stream cipher**, with particular focus on the **quarter-round function**. Some tasks require you to perform **manual calculations**, while others require you to **implement and experimentally evaluate** the algorithm using a programming language of your choice.

You are expected to complete the following types of tasks:

- **Conceptual Analysis Tasks:** Explain cryptographic concepts such as diffusion, modular addition, XOR operations, and bit rotations used in ChaCha20.
- **Manual Computation Tasks:** Perform step-by-step calculations of the ChaCha QuarterRound function using provided input values.
- **Implementation Tasks:** Implement the ChaCha quarter-round function in a programming language (e.g., Python, C, or Java) and demonstrate its correctness.

- **Experimental Analysis Tasks:** Conduct small experiments (such as flipping one bit of the input) and analyze how the output changes to demonstrate diffusion properties.

Because each student may choose different implementation approaches and experimental inputs, **your intermediate results, outputs, and analysis may differ**. This is expected and forms part of the evaluation.

## Submission Policy

- Students are allowed to submit this assignment **only once**.
- There will be **no second submission or resubmission opportunity**.
- Please ensure that your report, code, and all required materials are complete and correctly uploaded before the deadline.
- All submissions must be uploaded through **Canvas** before the assignment deadline.
- If you miss the submission deadline, make sure you put more effort for the next assignment
- **References** to sources used need to be explicitly mentioned at the end of the submission report

It is the student's responsibility to verify that the correct files have been submitted on Canvas. Incomplete or incorrect submissions may result in loss of marks according to the grading rubric.

## Academic Integrity and Collaboration

- **Allowed:** Discussing lecture material and cryptographic concepts at a high level.
- **Not allowed:** Sharing code, decrypted messages, keys, screenshots, or report content with other groups.
- **Not allowed:** Copying solutions from the internet, generative AI tools, previous students, or public repositories.

## Assignment Objectives

The objective of this assignment is to develop a deeper understanding of the **ChaCha20 quarter-round function**, which forms the fundamental building block of the ChaCha stream cipher. Through a combination of theoretical analysis, manual computation, implementation, and experimentation, students will explore how simple arithmetic and bitwise operations are used to achieve strong cryptographic diffusion.

By completing this assignment, students will be able to:

- Understand the role of the quarter-round transformation in the ChaCha cipher design.
- Explain the cryptographic significance of **modular addition**, **XOR operations**, and **bit rotations** used in ARX-based stream ciphers.
- Perform **manual step-by-step computations** of the ChaCha quarter-round function to better understand how the internal state evolves.
- Implement the ChaCha quarter-round function in a programming language and verify its correctness through experimental evaluation.
- Investigate the **diffusion properties** of the ChaCha design by conducting controlled experiments such as bit-flipping and observing avalanche effects.
- Develop analytical thinking skills by exploring how small changes in the algorithm or input parameters influence the behavior of the cipher.

## 1 Introduction

Modern stream ciphers are widely used to provide fast and secure encryption in many real-world systems, including network protocols, secure communications, and operating systems. One of the most widely deployed modern stream ciphers is **ChaCha20**, designed by Daniel J. Bernstein as a variant and improvement of the Salsa20 cipher family. ChaCha was introduced to improve the diffusion properties of Salsa20 while maintaining high performance and efficiency across a wide range of hardware platforms. Students are encouraged to consult the original paper by Daniel J. Bernstein, *ChaCha: A Variant of Salsa20* (2008), available at: <https://cr.yp.to/chacha/chacha-20080120.pdf>.

The ChaCha cipher operates on a  $4 \times 4$  matrix of 32-bit words and performs a sequence of transformations known as *rounds*. Each round applies a set of arithmetic and logical operations that mix the internal state to produce pseudorandom output. The core building block of this transformation is the **quarter-round function**, which updates four 32-bit state words using a combination of modular addition, XOR operations, and bit rotations. These simple operations are carefully arranged to achieve strong diffusion, ensuring that small changes in the input rapidly propagate throughout the state.

In his paper “*ChaCha, a Variant of Salsa20*”, Bernstein presents the ChaCha cipher and describes how modifications to the original Salsa20 design improve the amount of diffusion achieved per round while maintaining comparable computational cost. In particular, the ChaCha quarter-round updates each state word twice and uses a different sequence of rotation constants, allowing input bits to influence output bits more rapidly compared to Salsa20.

The goal of this assignment is to study and understand the ChaCha quarter-round function. You will analyze the structure of the algorithm, manually compute selected transformations, implement the function programmatically, and experimentally evaluate its diffusion properties. Through these tasks, it is expected that you will have some practical experience with fundamental cryptographic operations such as modular arithmetic, XOR operations, and bitwise rotations, which are commonly used in modern symmetric cryptographic designs.

## 2 ChaCha20 Quarter-Round Function

The fundamental operation in the ChaCha stream cipher is the **quarter-round function**. This function operates on four 32-bit state words and performs a sequence of arithmetic and bitwise transformations designed to mix the internal state efficiently. The quarter-round is a key component of the ChaCha round structure and is responsible for spreading input differences across multiple words in the state matrix.

ChaCha uses a combination of three basic operations that are efficient on modern processors:

- **Addition modulo  $2^{32}$**
- **Bitwise XOR**
- **Bit rotation**

These operations are commonly used in symmetric cryptographic designs because they are fast, simple to implement in hardware and software, and contribute to strong diffusion and confusion properties in cipher constructions.

According to Bernstein's ChaCha design, the quarter-round function updates four state words  $a$ ,  $b$ ,  $c$ , and  $d$  using the following sequence of operations:

```
a += b; d ^= a; d <<< 16;  
c += d; b ^= c; b <<< 12;  
a += b; d ^= a; d <<< 8;  
c += d; b ^= c; b <<< 7;
```

Here, the operator  $+$  denotes addition modulo  $2^{32}$ , the symbol  $\oplus$  (or  $\wedge$ ) represents the XOR operation, and  $\lll$  denotes a left rotation of a 32-bit word. These operations are arranged so that each input word influences every output word during the transformation. As a result, even small changes in the input values quickly propagate through the state, improving the diffusion properties of the cipher.

One important design improvement in ChaCha compared to Salsa20 is that each state word is updated twice during the quarter-round process. This allows input differences to spread more rapidly across the internal state. Bernstein notes that this structure increases the average number of output bits affected by a single input-bit change, thereby improving the diffusion per round while maintaining similar computational cost.

## 3 Assignment Tasks

In this assignment, you will analyze and implement the **ChaCha20 quarter-round function**. The tasks are designed to help you understand the internal operations of the cipher and observe how simple arithmetic and bitwise operations create strong diffusion in cryptographic algorithms.

### 3.1 Task 1: Understanding the Operations

Before performing the ChaCha transformation, briefly explain the following operations used in the quarter-round function:

- Addition modulo  $2^{32}$
- XOR operation
- Bit rotation (left rotation)

Explain why these operations are commonly used in modern symmetric cryptographic designs.

### 3.2 Task 2: Manual Quarter-Round Computation

In this task, you must manually compute the ChaCha quarter-round transformation using **pen and paper**. Show every intermediate step of your calculations.

You are given the following 32-bit input words:

$$a = 0x11111111, \quad b = 0x01020304, \quad c = 0x9b8d6f43, \quad d = 0x01234567$$

Apply the ChaCha quarter-round operations in the following order:

```
a += b; d ^= a; d <<< 16
c += d; b ^= c; b <<< 12
a += b; d ^= a; d <<< 8
c += d; b ^= c; b <<< 7
```

**Instructions:**

- Perform each operation step by step.
- Show intermediate values for  $a$ ,  $b$ ,  $c$ , and  $d$  after every operation.
- Clearly indicate the result after each modular addition, XOR operation, and rotation.

Your submission must include **all intermediate results** to demonstrate your understanding of the transformation.

### 3.3 Task 3: Implementation of the Quarter-Round

Implement the ChaCha quarter-round function in a programming language of your choice (e.g., Python, C, or Java).

Your program should:

- Accept four 32-bit integers as input
- Apply the ChaCha quarter-round operations

- Output the updated values of  $a$ ,  $b$ ,  $c$ , and  $d$

Use the same input values provided in Task 2 and verify that your program produces the same results.

Include your source code and a screenshot or output log showing the program execution and explain what happens in each step.

### 3.4 Task 4: Diffusion Experiment

To observe diffusion, repeat the quarter-round computation after flipping **one bit** in the input value  $a$ .

- Run the algorithm using the original inputs.
- Flip one bit in  $a$ .
- Run the algorithm again.

Report:

- the original output values
- the modified output values
- the number of bits that changed in the result

Briefly explain what this demonstrates about the diffusion properties of the ChaCha design.

### 3.5 Task 5: Critical Reflection

Answer the following questions:

- Why does ChaCha use multiple rotations (16, 12, 8, and 7)?
- How does the quarter-round contribute to diffusion in the cipher?
- Why are modular addition, XOR, and rotations effective together in ARX-based cryptographic designs?
- What is  $(a, b, c, \text{ and } d)$  and what are their influence?

## 4 Research Exploration Tasks

In the previous tasks, you manually computed and implemented the **ChaCha quarter-round function**. You also conducted a small diffusion experiment by flipping a single input bit and observing the resulting changes in the output.

In this section, you will extend your implementation and observations to explore deeper properties of the ChaCha design. These questions encourage you to think about how cryptographic primitives are evaluated and analyzed in research.

## Task 6: Avalanche Effect Analysis

Using the implementation you developed in **Task 3**, perform multiple avalanche experiments.

- Flip one bit in one of the input words ( $a$ ,  $b$ ,  $c$ , or  $d$ ).
- Run the quarter-round function again.
- Measure how many output bits change compared to the original output.

Repeat this experiment several times with different bit positions.

**Question:** What is the average number of bits that change in the output? Does the ChaCha quarter-round demonstrate strong diffusion based on your observations?

## Task 7: Rotation Constant Exploration

The ChaCha quarter-round uses the rotation constants:

16, 12, 8, 7

Modify your implementation by changing **one** of these rotation values (for example change 16 to 15 or 17).

Using the same experiment performed earlier:

- Run the diffusion experiment again.
- Compare the number of changed output bits with the original ChaCha design.

**Question:** How does changing the rotation constant affect diffusion? Why do you think the ChaCha designers carefully selected these specific values?

## Task 8: Input Word Influence Analysis

Using the same input values from **Task 2**, perform the following experiment:

- Flip one bit in  $a$  and record the output.
- Flip one bit in  $b$  and record the output.
- Flip one bit in  $c$  and record the output.
- Flip one bit in  $d$  and record the output.

**Question:**

Does a change in one input word affect all output words? Explain how the quarter-round structure allows input changes to propagate through the state.

## Task 9: Reduced Transformation Analysis

Your implementation currently performs a full quarter-round transformation. Modify your code so that you observe the intermediate outputs after:

- the first operation block
- the second operation block
- the third operation block
- the final operation block

### Question:

How does diffusion increase as the transformation progresses? Which stage of the quarter-round appears to contribute most to spreading the differences?

## Task 10: Reflection on ARX Cryptographic Design

ChaCha belongs to a family of ciphers known as **ARX ciphers**, which rely only on:

- modular addition
- bit rotation
- XOR operations

Based on your manual computation and implementation in this assignment:

### Question:

Why do you think these simple operations are effective when combined together in cryptographic design? Discuss how they contribute to confusion and diffusion.

## 5 Deliverables

Each student must submit the following items:

- **Written Report (PDF)**

Your report should clearly present your work for all tasks in the assignment. The report must include:

- Explanation of each step and each screenshot, failure will lead to a reduction of points
- Answers to the conceptual questions about the operations used in the ChaCha quarter-round.
- The complete **manual computation** of the quarter-round function showing all intermediate steps.

- Results and explanation of the **diffusion experiment**.
- Responses to the **research exploration questions**.

- **Manual Computation Work**

- Show all intermediate values for  $a$ ,  $b$ ,  $c$ , and  $d$  after each operation.
- Your work may be handwritten and scanned or clearly typed in the report.

- **Source Code**

- Implementation of the ChaCha quarter-round function in a programming language of your choice (e.g., Python, C, Java).
- The code must follow the algorithm presented in the assignment.
- Include comments explaining important steps in the implementation.

- **Program Output**

- Include screenshots or output logs demonstrating that your implementation runs correctly.
- Show the results produced for the input values provided in the assignment.

- **Experimental Results**

- Present the results of the avalanche/diffusion experiments.
- Clearly compare the outputs before and after modifying input bits or rotation constants.

All submissions must be uploaded through **Canvas** before the deadline.

All files must be submitted through **Canvas** before the deadline. Late submissions may be subject to penalties according to the course policy.

## 6 Grading Rubric

The assignment will be graded out of **100 points**. The breakdown below shows how points will be awarded.

| Component                                   | Assessment Criteria | Points |
|---------------------------------------------|---------------------|--------|
| <b>Task 1: Understanding the Operations</b> |                     |        |

| Component                                       | Assessment Criteria                                                                           | Points |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------|--------|
| Explanation of Modular Addition                 | Clear explanation of addition modulo $2^{32}$ and its importance in cryptographic algorithms. | 4      |
| Explanation of XOR Operation                    | Accurate description of XOR and its role in mixing bits in cryptographic functions.           | 3      |
| Explanation of Bit Rotation                     | Correct explanation of bit rotations and their contribution to diffusion.                     | 3      |
| <b>Task 2: Manual Quarter-Round Computation</b> |                                                                                               |        |
| Correct Modular Additions                       | All modular additions are computed correctly.                                                 | 10     |
| Correct XOR Operations                          | All XOR operations are correctly applied.                                                     | 5      |
| Correct Bit Rotations                           | Rotations by 16, 12, 8, and 7 bits are correctly performed.                                   | 5      |
| Intermediate Steps Shown                        | All intermediate values are clearly shown and logically organized.                            | 5      |
| Final Output Values                             | Final values of $a$ , $b$ , $c$ , and $d$ are correct.                                        | 5      |
| <b>Task 3: Implementation</b>                   |                                                                                               |        |
| Correct Algorithm Implementation                | Code correctly implements the ChaCha quarter-round algorithm.                                 | 12     |
| Code Functionality                              | Program runs successfully and produces expected outputs.                                      | 8      |
| Code Structure and Documentation                | Code is readable, well organized, and properly commented.                                     | 5      |
| Verification Against Manual Results             | Program output matches manual computation results.                                            | 5      |
| <b>Task 4: Diffusion Experiment</b>             |                                                                                               |        |
| Experiment Setup                                | Correctly modifies input bits and reruns the algorithm.                                       | 6      |
| Output Comparison                               | Original and modified outputs are clearly presented.                                          | 6      |
| Bit Difference Measurement                      | Correct calculation of the number of changed bits.                                            | 4      |
| Explanation of Diffusion                        | Clear explanation of how the results demonstrate diffusion.                                   | 4      |
| <b>Research Exploration Tasks</b>               |                                                                                               |        |
| Experimental Exploration                        | Student performs additional experiments (rotation modification or multiple avalanche tests).  | 6      |
| Analysis and Interpretation                     | Student provides thoughtful analysis and interpretation of results.                           | 6      |
| Connection to ChaCha Design                     | Explanation connects observations to ChaCha design principles and ARX cryptography.           | 8      |

| Component | Assessment Criteria | Points |
|-----------|---------------------|--------|
| Total     |                     | 100    |